

13	B	By Newton's 2nd law, F_{net} on piston = ma Since gas exerts an upward force $F_{\text{net}} = F_{\text{gas}} - mg$ where mg is the piston's weight Hence $F_{\text{net}} = F_{\text{gas}} - mg = ma$ $F_{\text{gas}} = m(g + a) = 0.75 (9.81 + 3.5) = 9.983 \text{ N}$ Work done by the gas = $F_{\text{gas}} \times s = 9.983 \times 0.12 = 1.20 \text{ J}$
14	B	Using $v_0 = \omega x_0$, $3 = \frac{2\pi}{12} x_0$ giving $x_0 = 5.730 \text{ cm}$ Using $a_0 = \omega^2 x_0$, $a_0 = \left(\frac{2\pi}{12}\right)^2 \times 5.730 = 1.571 \text{ cm s}^{-2}$ Hence maximum restoring force = $3.0 \times 1.571 \times 10^{-2} = 4.7 \times 10^{-2} \text{ N}$ Option C: wrong $T = 15/1.5 = 10 \text{ s}$ Option D: forgot to square omega, and mass multiply by max $v = 3.0 \times 0.03$
15	C	Since the wave pulse is travelling to the right, the last point on the displacement-time graph will be